东南大学考试卷（期中卷）

课程名称 操作系统原理及应用 考试学期 10-11-3 得分 84

适用专业 软件学院 考试形式 闭卷 考试时间长度 120分钟

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一、填空题（每空1分，共20分） 20

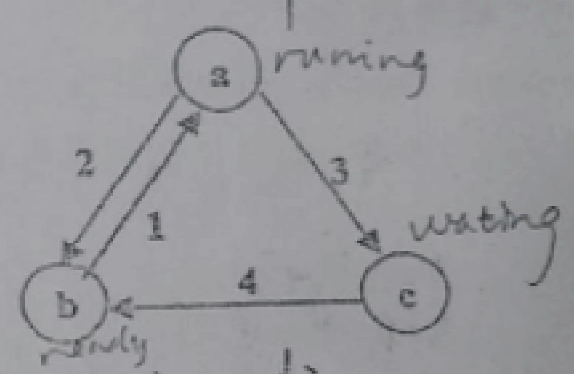
1. 为了确保操作系统的正常执行，计算机系统中程序的执行模式分为 用户态模式 和 核心态模式。

2. 操作系统提供给用户的接口是 图形界面接口 和 命令行接口，提供给应用程序的接口是 系统调用接口

3. 操作系统中共有三类调度器，它们是短期调度(short-term scheduler)，长期调度 和 中期调度；在一般操作系统中，必须具备的调度器是 短期调度器

4. Linux 操作系统中用户程序可以使用 fork 系统调用来创建一个子进程。

5. 某系统的进程状态(三态模型，无创建态和终止态)如右图所示：a 是 running运行 状态，b 是 ready就绪 状态，c 是 waiting等待 状态，其中，1 表示 CPU从就绪队列选一个进程运行，2 表示 发生中断，如超时，3 表示发生了等待事件，4 表示 等待事件完成。



6. 在一个单处理器系统中，存在5个进程，则最多有 4 个进程处于就绪状态，最多 5 个进程处于等待状态。

7. 进程调度算法采用固定时间片轮转法，时间片过大，就会使轮转法转化为 先来先服务 调度算法。 FCFS

8. 在现代操作系统中，资源的最小分配单位是 进程，而 CPU 的最小调度单位是 线程。

二、选择题（每题2分，共30分） 26

1. 采用多道程序设计技术能提高整个计算机系统的效率，其基本条件是（C） 2

A. 处理器执行指令速度快　　B. 硬盘容量大

C. 该系统具有处理器与外设并行工作的能力　　D. 外围设备多

2. 为了使操作系统中所有的用户都能得到及时的响应，该操作系统应该是（ B ） 2

A. 多道批处理系统　　B. 分时系统　　C. 实时系统　　D. 网络系统

3. 以下指令中，（ B ）不是特权指令。 2

A. I/O 指令　　B. 读取当前时钟　　C. 打开中断　　D. 关闭中断

4. 多个进程同时存在于内存中，并在一段时间内同时运行，这种性质称作（ B ）。2

A. 动态性　B. 并发性　C. 调度性　D. 异步性

5. 在操作系统中引入"进程"概念的主要目的是（ B ）。2

A. 改善用户编程环境　B. 描述程序动态执行过程的性质

C. 使程序与计算过程一一对应　D. 提高程序的运行速度

6. 进程控制块是描述进程状态和特性的数据结构，一个进程（ D ）。2

A. 可以有多个进程控制块　B. 可以和其他进程共用一个进程控制块

C. 可以没有进程控制块　D. 只能有惟一的进程控制块

7. 下列进程状态的转换中，哪一个是不正确的（ C ）。2

A. 就绪→运行　B. 运行→就绪　C. 就绪→阻塞　D. 阻塞→就绪

8. 进程所请求的一次打印输出结束后，将使进程状态从（ D 2

A、运行态变为就绪态　B、运行态变为等待态

C、就绪态变为运行态　D、等待态变为就绪态

9. 进程上下文中包含如下各项，除了（ C ）。2

A. 用户打开文件表　B. P C B　C. 中断向量　D. 内核栈

10. 下面步骤中，（ A ）不是创建进程所必需的。2

A. 由调度程序为进程分配 CPU　B. 建立一个进程控制块

C. 为进程分配内存　D. 将进程控制块链入就绪队列

11. 三个作业同时到达，运行时间分别为 T1、T2 和 T3，且 T1<T2<T3，若它们在单处理器系统中运行，采用短作业优先算法，则平均周转时间为（ D ）。2

T_1 T_2 T_3

A. T1+T2+T3　B. （T1+T2+T3）/3

C. T1+1/3*T2+2/3*T3　D. 1/3*T3+2/3*T2+T1

$\frac{T_1+T_1+T_2+T_1+T_2+T_3}{3}$

12. 关于 Round Robin 调度算法，以下说法正确的是（ C ）。2

A. 同样情况下，时间片越大，平均周转时间越小 ×

B. Round Robin 算法是 FCFS 算法的一种特殊情况 ×

C. 只有实现了定时器机制，才能实现 Round Robin 算法 ~~×~~ √

D. Round Robin 属于非抢占调度算法 √

13. 两个进程合作完成一个任务。在并发执行中，一个进程要等到其合作伙伴发来消息，或建立某条件后再向前执行，这种制约性合作关系被称为进程的（ A ）2

A. 同步　B. 互斥　C. 调度　D. 执行

14. 所谓临界区是指（ D / A） 0

A. 更改共享数据的一段程序　B. 一个共享变量

C. 一种同步机制　D. 访问临界资源的一段程序

15. 对于两个并发进程，设互斥信号量为 mutex，若 mutex=0，则（   ）

A. 表示没有进程进入临界区　　B. 表示有一个进程进入临界区

C. 表示有两个进程进入临界区　　D. 表示有一个进程进入临界区，另一个进程等待进入

三、简答题（共 15 分）

1. 简述 Linux 操作系统中中断机制与信号机制之间的异同。（6 分）
2. 简要说明程序、进程和线程之间的区别与联系。（5 分）
3. 进程调度中“可抢占”和“非抢占”两种方式哪一种系统开销更大？为什么？（4 分）

四、计算题（共 10 分）

假定在单 CPU 条件下有下列要执行的进程：

进程	运行时间	优先级
1	10	2
2	4	3
3	3	5

进程到达时间是按进程编号顺序进行的（即后面的进程依次比前一个进程迟到一个时间单位），优先级值越大代表优先级越高，

（1）用执行时间图（甘特图）描述在采用非抢占式优先级算法时的进程调度情况，并计算其平均周转时间和平均等待时间。

（2）如采用抢占式优先级算法，用执行时间图描述进程调度情况，并计算其平均周转时间和平均等待时间。

五、程序分析题（共 15 分）

1. 阅读以下程序段：

```
#include <pthread.h>
#include <stdio.h>
int value = 0;
void *runner(void *param){
    value = 5;
    pthread_exit(0);
}

int main(int argc, char *argv[])
{    int pid;
```

```
pid = fork();
if (pid == 0) {
   pthread_create (&tid, NULL, runner, NULL);
   pthread_join (tid, NULL);
   printf ("CHILD: value = %d", value);   /*Print 1*/
}
else if (pid > 0){
   wait(NULL);
   printf("PARENT: value = %d", value);   /*Print 2*/
}
}
```

5.

(1) 写出程序运行后的输出结果，并说明其原因。

(2) 分别指出 pthread_join()和 wait()的作用。(共 6 分)

2. 两个进程 P0 和 P1（如下所示，其中 i 取值为 0 或 1，j=1-i）共享两个数据项 int turn 和 boolean flag[2]，初始值为 turn=0，flag[0]=flag[1]=false。试从临界区问题解决方案必须满足的三个条件讨论下图所示算法能否解决 P0 和 P1 互斥访问临界区问题。(9 分)

```
Pi:  do {
        flag[i] = true;
        while (turn != i)
        {
           while (flag [j]);
           turn = i;
        }
        临界区
        flag [i] = false;
        剩余区
     } while (true)
```

P0 P1

不满足互斥性

P0先不进，P1先进去

while (turn != i)

{

while (flag[j]);

turn = i;

}

turn=0 flag[0]=flag[1]=false.

P0

flag[0]=true

~~turn~~ while.

P1

P0

P1 flag[1]=true

六、综合题（10 分）

假定一个阅览室最多可容纳 100 人，读者进入和离开阅览室时都必须在阅览室门口的一个登记表上登记，而且每次只允许一人进行登记操作。请使用信号量实现上述同步过程。

P1 flag[1]=true turn=0

turn=1

P0 flag[0]=true ↓ 临界区.

P1

2013 级东南大学操作系统期中考试试卷

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Classify each statement as true or false by ticking the correct box. [5×2 pts]

[T] [F] Modern operating systems are interrupt driven.

[T] [F] I/O instructions and the instructions that turn interrupts on are generally considered to be privileged instructions.

[T] [F] Threads are cheaper to create than processes.

[T] [F] Kernel-scheduled threads are cheaper to create than user-level threads.

[T] [F] Shortest-Job-First scheduling algorithm is one of the usual algorithms used for CPU scheduling in modern operating systems.

2: Short Answer questions [7×5 pts] (Answer all the questions on the *answer sheet*.)

2. Short Answer questions [7×5 pts] (Answer all the questions on the *answer sheet*.)

(a) Describe the differences between process and program.

(b) What are the differences between I/O bound and CPU bound processes? Describe one typical application for I/O bound processes and CPU bound processes, respectively.

(c) Draw the five state transition diagram of a process and mark the corresponding event that might cause some transition.

(d) Describe the benefits of multithreaded programming. Give an example of practical problem whose multithreaded solution provides better performance than its single-threaded solution on a single-processor system.

(e) List and explain criteria we might use to evaluate a CPU scheduling algorithm.

(f) Describe the differences among long-term, medium-term and short-term process scheduling.

(g) Give the definition of busy waiting. Under what conditions might it be desirable and undesirable respectively?

3. CPU Scheduling [15 pts] (Answer this question on the *answer sheet.*)
Consider the following set of processes that arrive at different times with the length of the CPU busrt time given in milliseconds:

Process	Arrival Time	Burst Time
P1	0	17
P2	12	8
P3	4	13
P4	7	7
P5	2	14

(a) Draw a complete Gantt Chart to illustrate the execution of the processes using Preemptive Shortest Job First scheduling algorithm.

(b) Show calculation of Average Waiting Time.

4. CPU Scheduling [15 pts] (Answer this question on the *answer sheet.*)
Consider a multi-level feedback queue in a single-CPU system. The first level (queue 0) is given a quantum of 8 ms, the second one a quantum of 16 ms, the third is scheduled FCFS. Assume jobs arrive all at time zero with the following job times (in ms): 4, 7, 12, 20, 25 and 30, respectively.

(a) Draw a Complete Gantt Chart to illustrate the execution of the processes using the multi-level feedback queue scheduling described above.

(b) Show calculation of Average Waiting Time.

5. Process Synchronization [15 pts] (Answer this question on the *answer sheet.*)
(a) State the three *correctness properties* to a solution of the critical section problem. Give each of the properties a short explanation.

(b) Peterson's algorithm to solve the critical section problem for two processes was adopted by a Student as below, messing up some of the variables. With respect to the three *correctness properties*, indicate which one is correct and which one is incorrect, and give your reasons.

6. Process Synchronization [10 pts] (Answer this question on the *answer sheet.*)
Give a pseudo-code solution to the readers/writers problem in which writers are favored (specifically, if a writer is waiting to start writing, then no new readers are allowed to start reading). Use semaphores as your synchronization primitive.

老师讲解笔记：

综合是简答题、 算法、调度时间公平。

判 十简 + 计算题。

期中：
判）①
②I/O指令和修改中断向量指令一般是特权指令(√)。
③创建一个线程开销小于进程。(√)
线程结构简单些。
④ 内核级线程开销=用户 — — (×)
⑤ SJF 只适用于批处理作业中，
最常使用算法之一 (×)

简：
(1) ① 静态永久。 临时动态
② 映射关系不一一对应 进程→多程序
③ 存在的依赖性，进程不可脱离程序存在

(2)
T(I/O 操作) > CPU Time I/O bound 打印机进程
< cpu bound. 科学计算、大数据处理

(3)
进程三状态图、阻塞事件、条件

(4) 多线程编程优点：①响应性
②可实现资源的共享。
③经济效益高，线程的创建开销小于进程
上下文切换
④~~多处理器~~ 并发性好。
多处理器程序中，适合多线程编程

例子：一个web处理程序中，监听线程、处理线程，提高并发性。

中 → 由内存中移到磁盘中后再移进来

长 → 进程从 buffer 中移到内存中

(7). 忙等? 进行等待资源，一直检测需要的资源

忙等在什么情况下 是有意义的

优点：进程一直占有 CPU 而不进行上下文切换，上下文切换可能需要更长 time.

在单处理器系统中，忙等是~~[illegible]~~不好的

多处理器中 — — — — 好的.

临界区的等待和进程状态的等待不一样.
可能在 running、waiting 状态.

~~[illegible]~~ 时

P_4 2 不抢占.

3. (a)

P_1	P_5	P_4	P_2	P_5	P_3	P_1

0　2　7 (4)　14 (12)　22　31　44　59

7 (4) 不抢占

P_1 15　P_5 9　P_3 13　P_4 7 (抢占)

判断剩下的是否抢占 (15 > 14)

SJFS 最短作业优先

在到达时间后判断哪个最小? 会不会抢占?

P_1 (平均等待时间) = 59 − 0 − 17 = 42
(结束时间) (开始) (执行时间)

P_2 (— —) = 22 − 12 − 8 = 2
(结束) (开始) (执行)

P_3 (— —) = 44 − 4 − 13 = 27

= 14 − 7 − 7 = 0

P5 | P6

0 4 11 19 27 35 43 47 59 75 91 92 98

怎么做？按顺序（默认了）　P3. P4 P5 P6 在第一个时间不能执行完

↓

下到第二个时间　P3 P4 做完

P5 P6 未完

↓

下到第三个时间

P5 剩1个时间

P6 剩6个

(b) Average waiting Time

$P_1 = 4-4=0$

$P_2 = \cancel{4}\ 11-7=4$

$P_3 = 47-12=35$

$P_4 = 59-20=39$

$P_5 = 92-25=67$

$P_6 = 98-31=68$

$$\frac{0+4+35+39+67+68}{6} = 35.5$$

（考试除不尽就写分数）

5.(a)
- 互斥：在任一个时间，不能有2个及以上进程对同一临界区
- 进展性
- 有限等待：当一个进程提出对一个临界区访问时，等待时间是有限的

(b) flag[2] and turn==2　轮到自己，互斥区访问.

互斥性不满足. P_1 进后，P_2 想进是可以进的.

P_1 想进就可以进.　P_2 也可进　在同时间 P_1, P_2 2个都要进都不进。

turn 只能为1或2.

东 南 大 学 考 试 卷（ A卷）

课程名称	操作系统原理	考试学期	05-06	得分	
适用专业	计算机	考试形式	闭卷	考试时间长度	120 分钟

Ⅰ Explain the following clauses in the course of OS.(7.5×4=30 points)

1. SPOOLING
2. System Call
3. multi-programming
4. process deadlock

Ⅱ Answer the following questions.(10+15=25 points)

1. Why it is necessary to use an *open* file command before to *read* (or *write*) a file?
2. Which kinds of interface commands are provided by device-driver to the kernel I/O subsystem?

Ⅲ Computation(18 points)

Consider the following page-reference string:

1,2,3,4,2,1,5,6,2,1,2,3,7,6,3,2,1,2,3,6

How many pages faults would occur for the following replacement algorithms, assuming three frames?

Remember that all frames are initially empty, so your first unique pages will all cost one fault each.

1.LRU replacement

2.FIFO replacement

3.Optimal replacement

Ⅳ Programming(27 points)

The solution to the first readers-writers problem does not prevent waiting writers from starving: If the database is currently being read and there is a waiting writer, subsequent readers will be allowed to read the database before the writer can write. Modify the solution such that it does not starve waiting writers.

1.Give a modification approach and

2.Write a segment of java code to implement this approach

东 南 大 学 考 试 卷（ B 卷）

课程名称	操作系统原理	考试学期	05-06	得分	
适用专业	计算机	考试形式	闭卷	考试时间长度	120 分钟

Ⅰ Explain the following clauses in the course of OS.(7.5×5=30 points)

1. working-set
2. thrashing
3. process deadlock
4. critical section

Ⅱ Answer the following questions.(10+15=25 points)

5. Why should the bit map for file allocation be kept on mass storage, rather than in main memory?(10 points)
   Bit vector 用于存储空闲的存储空间位置，如果放在内存中，掉电后信息将会丢失，显然这一信息需要持久化存储。
6. State briefly the typical life cycle of an I/O request.(15 points)
   I/O Request —System Call->
   If the data is already in the buffer cache, this system call is returned.
   Otherwise, system sends a request to the device driver, progress is suspended (Waiting state). After the device driver finished the I/O manipulation, it will send an interrupt to the kernel. Kernel acknowledged that and set the I/O interrupt indicating I/O is finished. The progress will be placed in the ready queue, waiting the scheduler to assign it to the running queue. When his turn comes, the progress will resume and continues execution.

Ⅲ Computation(18 points)

A certain computer provides its users with a virtual-memory space of 2^{32} bytes. The computer has 2^{18} bytes of physical memory. The virtual memory is implemented by paging, and the page size is 4,096 bytes. A user process generates the virtual address 11123456.Explain how the system establishes the corresponding physical location. .

4096 bytes = 2^{12} bytes.

Address = | p: 20bits | d: 12 bits|

对于逻辑地址 1112 3456h 来说， 1 1123h 为页号， 456h 为页内地址偏移量

Ⅳ Programming(27 points)

The bounded-buffer producer/consumer problem is a classical synchronization paradigm in operating system.

Write a **BoundedBuffer** class in which the synchronized methods **enter (object item)** and **object remove()** are implemented. Assume that **wait()** and **notify()** methods are APIs available to programmers.

东 南 大 学 考 试 卷（B 卷）

课程名称 操作系统 考试学期 07-08-2
适用专业 计算机应用 考试形式 闭卷 考试时

Give the technical term that best fits these definitions
1. Critical Section
2. Prefetching
3. sppoling
4. Dynamic loading
5. Thrashing
6. Microkernel

自觉遵守考场纪律 如考试作弊 此答卷无效

一、Definitions [5X6=30 pts]

Give the technical term that best fits these definitions

1. Portion of a program that accesses shared variables and that no two processes can be executing this code that manipulates shared variables at the same time.

2. *It* is a method of overlapping the I/O of a job with that job's own computation. The idea is simple. After a read operation completes and the job is about to start operating on the data, the input device is instructed to begin the next read immediately. The CPU and input device are then both busy.

3. Some devices, such as tape drives and printers, cannot usefully multiplex the I/0 requests of multiple concurrent applications. The subsystems can coordinate concurrent output to a separate disk file. For instance, When an application finishes printing, the subsystem copies the queued files to the printer one, at a time.

4. Unused routine is never loaded.

5. A process is busy swapping pages in and out.

6. A small operating system core that provides basic scheduling, memory management and communication services while relying on processes to perform the other required functionality traditionally associated with the operating system.

二、Comparisons[6+9=15pts]

Complete the following comparisons using True, False or Possible

1.[6pts]Blocking I/o versus Noblocking I/O

Point of Comparison	Blocking I/O	Noblocking I/O
I/O call returns as much as available		
Implemented via multi-threading		

2. [9 pts] Threads versus processes

Point of Comparison	Threads	Processes
Share all global variables		
Execute in separate address spaces		
Require less time to create, terminate and switch from one to another		

三、Computing[10+12=22 pts]

1. Suppose that a disk drive has 5000 cylinders, numbered 0 to 4999. The drive is currently serving a request at cylinder 143, and the previous request was at cylinder 125. The queue of pending requests, in FIFO order, is

86, 1470, 913, 1774, 948, 1509, 1022, 1750, 130

Starting from the current head position, what is the total distance (in cylinders) that the disk arm moves to satisfy all the pending requests, for each of the following disk-scheduling algorithms? [10 pts]

a. SSTF

b. LOOK

2. Consider the following set of processes, with the length of the CPU-burst time given in milliseconds:

Process	Burst Time	Priority
$P1$	10	3
$P2$	1	1
$P3$	2	3
$P4$	1	4
$P5$	5	2

The processes are assumed to have arrived in the order $P1$, $P2$, $P3$, $P4$, $P5$, all at time 0. [12 pts]

a. Draw the Gantt chart illustrating the execution of these processes using SJF, a nonpreemptive priority (a smaller priority number implies a higher priority) scheduling.

b. What is the turnaround time of each process for the scheduling algorithm in part a?

c. What is the waiting time of each process for the scheduling algorithm in part a?

四、Answer the following questions [8+10=18 pts]

1.What is a race condition? When does it happen? [8pts]

2. Under what circumstances do *page faults* occur ? Describe the actions taken by the operating system when a page fault occurs. [10pts]

五、Programming[15 pts]

The Sleeping-Barber Problem. A barbershop consists of a waiting room with n chairs, and a barber room with one barber chair. If there are no customers to be served, the barber goes to sleep. If a customer enters the barbershop and all chairs are occupied, then the customer leaves the shop. If the barber is busy, but chairs are available, then the customer sits in one of the free chairs. If the barber is asleep, the customer wakes up the barber. Write a program to coordinate the barber and the customers using Java synchronization（or vc++ with P and V operations）.

B 卷答案

一、Definitions [5X6=30 points]

Give the technical term that best fits these definitions

1. Critical Section
2. Prefetching
3. sppoling
4. Dynamic loading
5. Thrashing
6. Microkernel

二、Comparisons[15pts]

Complete the following comparisons using True, False or Possible

1)[6pts]Blocking I/o versus Noblocking I/O

False True

Possible True

2) [9 pts] Threads versus processes

True False

False True

True False

三、Computing[22 points]

1.

a.The SSTF schedule is

143, 130, 86, 913, 948, 1022, 1470, 1509, 1750, 1774.

The total seek distance is 1745.

b. The LOOK schedule is

143, 913, 948, 1022, 1470, 1509, 1750, 1774, 130, 86.

The total seek distance is 3319.

2.

Answer:

a. The two Gantt charts are

2 3 4 5 1 5 1 5 1 5 1 1 3 1 5

1 3 4 5 2

b. Turnaround time(SJF and priority)

P1 19 ,16 P2 1 ,1 P3 4 ,18 P4 2 ,19 P5 9 ,6

c. Waiting time (turnaround time minus burst time)

P1 9 ,6 P2 0 , 0 P3 2 ,16 P4 1 ,18 P5 4 ,1

四、Answer the following questions

1. The situation where several processes access – and manipulate shared data concurrently. The final value of the shared data depends upon which process finishes last.

2. During address translation, if valid–invalid bit in page table entry is 0 ⇒ page fault.

- If there is ever a reference to a page, first reference will trap to OS ⇒ page fault
- OS looks at another table(in PCB) to decide:
  - Invalid reference ⇒ abort.
  - Just not in memory ⇒page it in.
- Get empty frame.
- Swap page into frame.
- Reset tables, validation bit = 1.
- Restart instruction: Least Recently Used .